

SBLToolkit.jl & Generalized Similarity Profile Theory (GSPT) Core Math

Mathematical Framework for Stable Boundary Layer (SBL) Transitions and Bifurcation Gating

Partitions observed vertical profile curvature into intrinsic thermodynamic and coordinate-mapping components.

- **Richardson Mapping Function** $R(\zeta)$:

$$R(\zeta) = \zeta \frac{\phi_h(\zeta)}{\phi_m^2(\zeta)} = \zeta \frac{S_m^{3/2}(\zeta)}{S_h(\zeta)}$$

Reconciled under $S_m = \phi_m^{-2}$, $S_h = (\phi_m \phi_h)^{-1}$.

- **Height-Resolved Curvature Decomposition:**

$$Ri_{zz} = \underbrace{R''(\zeta)\zeta_z^2}_{C_{\text{const}}} + \underbrace{R'(\zeta)\zeta_{zz}}_{C_{\text{coord}}} + \mathcal{E}_{\Delta z}$$

- **Coordinate Jacobian** ζ_z :

$$\zeta_z = \frac{L(z) - zL'(z)}{L(z)^2} = \frac{1 - \zeta L'(z)}{L(z)}$$

- **Linear Obukhov Test (No-Fold Gate):** If $L(z) = L_0 + az$, $L_0 > 0$:

$$\zeta_z = \frac{L_0}{(L_0 + az)^2} > 0 \quad \forall z \geq 0$$

No finite coordinate turning points can exist.

- **Nondegenerate Fold Condition:** At a fold locus $z^* > 0$, $L(z^*) \neq 0$:

$$\zeta_z(z^*) = 0 \iff L(z^*) = z^* L'(z^*), \quad L''(z^*) \neq 0$$

$$\zeta_{zz}(z^*) = -\frac{z^* L''(z^*)}{L(z^*)^2}$$

- **Continuum Curvature at Fold:**

$$Ri_{zz}(z^*) = R'(\zeta^*)\zeta_{zz}(z^*) + \mathcal{E}_{\Delta z}$$

Strictly coordinate-driven at the fold.

Models SBL transitions as coupled microscale TKE (fast) and mesoscale shear (slow) dynamics.

Decoupling Principle & Operational Gate

$$\underbrace{\zeta_z = 0}_{\text{Kinematic Fold}} \not\iff \underbrace{\lambda_f = 0}_{\text{Bifurcation}}$$

$$\boxed{G = \mathbf{1} \ (\lambda_f \geq -\epsilon_\lambda)}$$

- **Fast Subsystem TKE Equation** ($\dot{e}$):

$$\epsilon \frac{de}{dt} = l_0 \sqrt{e + \delta} S^2 - B_{0,\text{max}} \left(\frac{e^2}{e^2 + \delta_{\text{reg}}^2} \right) - \frac{e^{3/2}}{l_0}$$

$$\delta = 10^{-6} \text{ m}^2 \text{ s}^{-2}; \quad B_{0,\text{max}} \text{ in } \text{m}^2 \text{ s}^{-3}.$$

- **Slow Subsystem Wind Shear** ($\dot{S}$):

$$\frac{dS}{dt} = G_0 - \gamma_s e S - r_s S$$

- **Fast Stability Eigenvalue** (λ_f):

$$\lambda_f(e, S) = \frac{1}{\epsilon} \frac{\partial F}{\partial e} \quad (\text{attracting if } \lambda_f < 0)$$

- **Saddle-Node Fold Locus:**

$$e_{\text{fold}} = \frac{l_0}{\sqrt{3}} \sqrt{S_{\text{fold}}^2 + \beta N^2}$$

Derived at the critical boundary $\partial F / \partial e = 0$.

Couples surface energy balance (SEB) directly into the tridiagonal vertical diffusion solver.

- **Surface Energy Balance:**

$$C_s \frac{\partial T_s}{\partial t} = R_n(t) - H_0(t) - G_s(t)$$

$$H_0 = \rho c_p K_h(z_1) \left(\frac{\theta_1 - T_s}{z_1} \right)$$

$$G_s = K_s (T_s - T_{\text{deep}})$$

- **Backward-Euler SEB Solution:**

$$\gamma \equiv \frac{\rho c_p K_h^{n+1}}{z_1}, \quad \beta \equiv \frac{C_s}{\Delta t} + \gamma + K_s$$

- **Updated Surface Temperature** (T_s^{n+1}):

$$T_s = \frac{1}{\beta} \left[\frac{C_s}{\Delta t} T_s^n + R_n^{n+1} + \gamma \theta_1^{n+1} + K_s T_{\text{deep}} \right]$$

- **Implementation:** Substituting T_s^{n+1} into the level-1 diffusion equation absorbs coupling into tridiagonal coefficients B_1, C_1 .

SBL Inversion Reference, Closures, and NWP Mandates

Regularization Standards, Similarity Closures, and Multi-Campaign Audit Guidelines

Smooths sensors prior to taking non-linear spatial quotients to bypass the non-commutation hazard.

- **The Non-Commutation Hazard:**

$$\mathcal{M}_\delta [A/B] \neq \mathcal{M}_\delta A / \mathcal{M}_\delta B$$

Filtering Ri_g manufactures false knees.

- **Tikhonov-Morozov Spline Functional:**

$$\min_{f_s} \sum_{i=1}^{N_{\text{obs}}} \left(\frac{f_s(\xi_i) - q_i}{\sigma_i} \right)^2 + \alpha \int_0^{\xi_{\text{max}}} [f_s''(\xi)]^2 d\xi$$

$\xi \equiv \ln(z/z_0)$. Solve α via bisection until

$$\|f_s(\xi_i) - q_i\|_2 = \delta$$

- **Tangential Cone Condition:**

$$\begin{aligned} \|F(x_2) - F(x_1) - F'(x_1)(x_2 - x_1)\| \\ \leq \gamma \|F(x_2) - F(x_1)\| \end{aligned}$$

Guarantees stable convergence of the MDP filter.

- **Analytical Derivatives:**

$$\begin{aligned} \frac{df}{dz} &= \frac{1}{z} \frac{df}{d\xi} \\ \frac{d^2 f}{dz^2} &= \frac{1}{z^2} \frac{d^2 f}{d\xi^2} - \frac{1}{z^2} \frac{df}{d\xi} \end{aligned}$$

Eddy diffusivity stability functions S_m, S_h .

- **Stability Function Definitions:**

$$\begin{aligned} K_m &= l_m^2 s S_m(\zeta), \quad K_h = l_m^2 s S_h(\zeta) \\ S_m(\zeta) &= \phi_m^{-2}(\zeta), \quad S_h(\zeta) = (\phi_m(\zeta) \phi_h(\zeta))^{-1} \end{aligned}$$

- **Symmetric Quadratic Closures ($Pr = 1$):**

$$S(Ri) = \begin{cases} (1 - Ri/Ri_c)^2, & Ri \leq Ri_c \\ 0, & Ri > Ri_c \end{cases}$$

$S(Ri), S'(Ri)$ are C^1 -continuous across $Ri_c = 0.25$.

- **GLGS (eMOST / SHEBA Formulation):**

$$\begin{aligned} \phi_m(\zeta) &= 1 + \frac{a_m \zeta}{(1 + b_m \zeta)^{2/3}} \\ \phi_h(\zeta) &= Pr_0 \left(1 + \frac{a_h \zeta}{1 + b_h \zeta} \right) \\ \psi_m(\zeta) &= -3 \frac{a_m}{b_m} \left[(1 + b_m \zeta)^{1/3} - 1 \right] \end{aligned}$$

- **Triple-Point Dispersion (δ_{TP}):**

$$\delta_{\text{TP}}(t) = \frac{\max(z_K, z_e, z_{e_z}) - \min(z_K, z_e, z_{e_z})}{H_{\text{SBL}}(t)}$$

Verified collapse events require $\delta_{\text{TP}} < 0.10$.

SBL Campaign Resolution Guidelines

Campaign	N_z	Feature	Status
CASES-99	150	LLJ Nose Fold	Success
GABLS3	150	Shear Erosion	SCM Closed
SHEBA	2	Stencil Collapse	Fallback

The Four Operational NWP Mandates:

1. **Track A Regularization:** Smooth primitive state vectors (u, v, θ_v) before computing Ri_g or spatial derivatives.
2. **Fast Stability Gating:** Gate transitions by direct eigenvalue evaluation $G = \mathbf{1}(\lambda_f \geq -\epsilon_\lambda)$ to prevent "Fold Illusion" shutdowns.
3. **Cease Empirical Tuning:** Do not adjust β_m, β_h to fit profile knees; these are kinematic coordinate features.
4. **Enforce $N_z \geq 3$ Ingestion Gates:** Bypass curvature stencils on sparse grids. Use Bulk Richardson fallback:

$$Ri_b = \frac{g}{\theta_v} \frac{(z_2 - z_1)(\theta_{v,2} - \theta_{v,1})}{(u_2 - u_1)^2 + (v_2 - v_1)^2}$$